

Lee, Jung-Hee

professor

基本信息

姓名: Lee, Jung-Hee

所属院校: 庆北国立大学

地区: South Korea

年龄范围: 50岁以上

海外经历: 有

学科领域

微电子科学与工程

集成电路设计与集成系统

电子科学与技术

研究领域

微电子器件与制造

纳米电子与先进工艺

半导体器件物理与工艺

半导体与光电子材料

模拟与射频IC设计

射频集成电路

宽禁带半导体器件

低功耗电路与能量管理

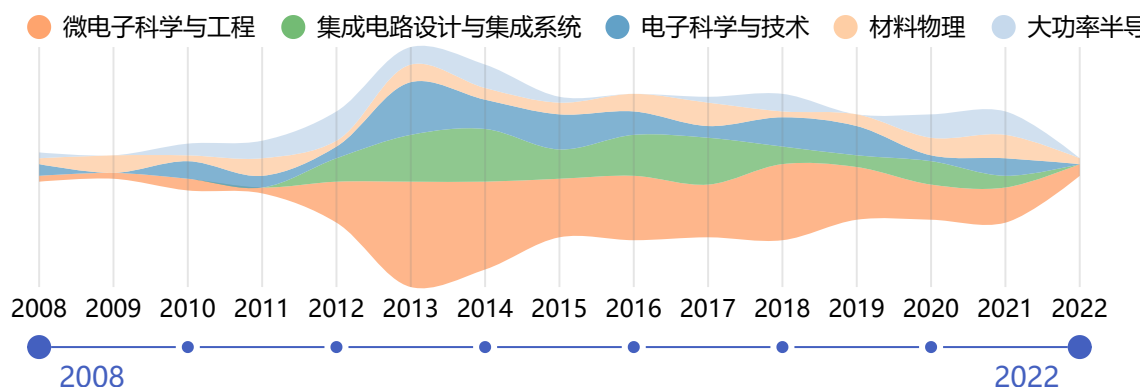
薄膜材料与表面物理

微纳电子与纳米技术

个人简介

Lee, Jung-Hee is a professor at the School of Electronics Engineering, Kyungpook National University. He leads the Nitride Semiconductor Device Laboratory, located at IT3 - 302 College of IT Engineering. His email is jleeee.knu.ac.kr and his phone is +82 - 53 - 950 - 6555. He holds a Ph.D. from North Carolina State University, USA. He has had various leadership roles at the university since 2008. His research interests include MOCVD, nitride - based compound semiconductors, LEDs, ALD, and semiconductor device physics.

近年发表的论文



Temperature dependence of the piezotronic and piezophototronic effects in flexible GaN thin films

NANO ENERGY

Mallem, Siva Pratap Reddy; Shim, Jaesool; Lee, Jung-Hee

被引用次数: 8

2022

Investigation of Proton Irradiation-Enhanced Device Performances in AlGaIn/GaN HEMTs

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Kim, Jeong-Gil; Kim, Eunjin; Kim, Dong-Seok; Cho, Chuyoung; Lee, Jung-Hee

被引用次数: 6

2022

Effect of Proton Irradiation Energy on AlGaIn/GaN HEMTs fabricated by Ion-implanted Isolation

2019 19TH EUROPEAN CONFERENCE ON RADIATION AND ITS EFFECTS ON COMPONENTS AND SYSTEMS (RADECS)

Kim, Dong-Seok; Lee, Jun-Hyeok; Kim, Jeong-Gil; Lee, Jae S.; Lee, Jung-Hee

被引用次数: 0

2022

Enhanced Stability and Sensitivity of AlGaIn/GaN-HEMTs pH Sensor by Reference Device

IEEE SENSORS JOURNAL

Dong, Yan; Wang, Rui; Xie, Zili; Liu, Yanli; Lei, Jianming; Guo, Hui; Dai, Quan; Kim, Jeong-Gil; Kang, Seung-Hyeon; Won, Chul-Ho; Lee, Jung-Hee; Chen, Dunjun; Zhang, Rong; Zheng, Youdou

被引用次数: 6

2021

Effect of gate dielectrics on characteristics of high-energy proton-irradiated AlGaIn/GaN MISHEMTs

RADIATION PHYSICS AND CHEMISTRY

Lee, Jun-Hyeok; Kim, Dong-Seok; Kim, Jeong-Gil; Ahn, Woo-Hyun; Bae, Youngho; Lee, Jung-Hee

被引用次数: 12

2021

Fabrication and characterization of GaN-based nanostructure field effect transistors

SOLID-STATE ELECTRONICS

Son, Dong-Hyeok; Thingujam, Terirama; Dai, Quan; Kim, Jeong-Gil; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 3

2021

Preface

SOLID-STATE ELECTRONICS

Lee, Jung-Hee; Kwon, Soon-Yong; Lee, Hi-Deok; Song, Taigon

被引用次数: 0

2021

Suppression of current dispersion in AlGa_N/Ga_N MISHFETs with in-situ Al_N passivation layer

SOLID-STATE ELECTRONICS

Lee, Jun-Hyeok; Kim, Jeong-Gil; Kang, Hee-Sung; Lee, Jung-Hee

被引用次数: 2

2021

Mechanism of Proton-Induced electrical degradation of AlGa_N/Ga_N high electron mobility transistors

SOLID-STATE ELECTRONICS

Kim, Dong-Seok; Kim, Jeong-Gil; Lee, Jun-Hyeok; Hwang, Yong Seok; Yoon, Young Jun; Lee, Jae Sang; Bae, Youngho; Lee, Jung-Hee

被引用次数: 3

2021

Strain-induced piezotronic effects in nano-sized Ga_N thin films

NANO ENERGY

Puneetha, Peddathimula; Mallem, Siva Pratap Reddy; Lee, Young-Woong; Lee, Jung-Hee; Shim, Jaesool

被引用次数: 14

2021

Temperature dependence of the piezotronic effect in CdS nanospheres

NANO ENERGY

Puneetha, Peddathimula; Mallem, Siva Pratap Reddy; Bathalavaram, Poornaprakash; Lee, Jung-Hee; Shim, Jaesool

被引用次数: 18

2021

Effects of Al Composition and High-Temperature Atomic Layer-Deposited Al₂O₃ Layer on the Leakage Current Characteristics of AlGa_N/Ga_N Schottky Barrier Diodes

CRYSTALS

Lee, Jae-Hoon; Lee, Jung-Hee; Im, Ki-Sik

被引用次数: 2

2021

A Simulation Study on the Effects of Interface Charges and Geometry on Vertical Ga_N Nanowire MOSFET for Low-Power Application

IEEE ACCESS

Thingujam, Terirama; Dai, Quan; Kim, Eunjin; Lee, Jung-Hee

被引用次数: 6

2021

High Breakdown Voltage and Low-Current Dispersion in AlGa_N/Ga_N HEMTs With High-Quality Al_N Buffer Layer

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Jeong-Gil; Cho, Chuyoung; Kim, Eunjin; Hwang, Jae Seok; Park, Kyung-Ho; Lee, Jung-Hee

被引用次数: 50

2021

Design optimization of GaN diode with p-GaN multi-well structure for high-efficiency betavoltaic cell

NUCLEAR ENGINEERING AND TECHNOLOGY

Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Jung-Hee; Kim, Dong-Seok

被引用次数: 3

2021

Effect of In-Situ Silicon Carbon Nitride (SiCN) Cap Layer on Performances of AlGaIn/GaN MISHFETs

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Lee, Jae-Hoon; Im, Ki-Sik; Lee, Jung-Hee

被引用次数: 2

2021

Experimental and simulation study of power performance improvement of GaN PIN betavoltaic cell

INTERNATIONAL JOURNAL OF ENERGY RESEARCH

Kim, Dong-Seok; Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Jung-Hee

被引用次数: 7

2021

Design and optimization of GaN-based betavoltaic cell for enhanced output power density

INTERNATIONAL JOURNAL OF ENERGY RESEARCH

Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Jung-Hee; Kim, Dong-Seok

被引用次数: 8

2021

Effects of Proton Irradiation on the Current Characteristics of SiN-Passivated AlGaIn/GaN MIS-HEMTs Using a TMAH-Based Surface Pre-Treatment

MICROMACHINES

Yoon, Young Jun; Lee, Jae Sang; Suk, Jae Kwon; Kang, In Man; Lee, Jung Hee; Lee, Eun Je; Kim, Dong Seok

被引用次数: 7

2021

AlInGaIn/GaN double-channel FinFET with high on-current and negligible current collapse

SOLID-STATE ELECTRONICS

Lee, Jun-Hyeok; Kim, Jeong-Gil; Ju, Jeong-Min; Ahn, Woo-Hyun; Kang, Seung-Hyeon; Lee, Jung-Hee

被引用次数: 8

2020

Leakage current and rectification behavior of Au / TiO₂ / GaN junctions with TiO₂ interlayer oxygen deposition pressure

CURRENT APPLIED PHYSICS

Lee, Youngjin; Kang, Seung-Hyeon; Lee, Jung-Hee; Dho, Joonghoe

被引用次数: 6

2020

Comprehensive Research of Total Ionizing Dose Effects in GaN-Based MIS-HEMTs Using Extremely Thin Gate Dielectric Layer

NANOMATERIALS

Chang, Sung-Jae; Kim, Dong-Seok; Kim, Tae-Woo; Lee, Jung-Hee; Bae, Youngho; Jung, Hyun-Wook; Kang, Soo Cheol; Kim, Haecheon; Noh, Youn-Sub; Lee, Sang-Heung; Kim, Seong-Il; Ahn, Ho-Kyun; Lim, Jong-Won

Gate Architecture Effects on the Gate Leakage Characteristics of GaN Wrap-gate Nanowire Transistors

ELECTRONIC MATERIALS LETTERS

Mallem, Siva Pratap Reddy; Im, Ki-Sik; Thingujam, Terirama; Lee, Jung-Hee; Caulmilone, Raphael; Cristoloveanu, Sorin

被引用次数: 2

2020

Influence of Thermal Annealing on the PdAl/Au Metal Stack Ohmic Contacts to p-AlGaN CRYSTALS

Mallem, Siva Pratap Reddy; Ahn, Woo-Hyun; Lee, Jung-Hee; Im, Ki-Sik

被引用次数: 3

2020

Anomalous DC Characteristics of AlGaIn/GaN HEMTs Depending on Proton Irradiation Energies

ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY

Kim, Dong-Seok; Lee, Jun-Hyeok; Kim, Jeong-Gil; Yoon, Young Jun; Lee, Jae Sang; Lee, Jung-Hee

被引用次数: 9

2020

Effect of Gate Structure on the Trapping Behavior of GaN Junctionless FinFETs

IEEE ELECTRON DEVICE LETTERS

Im, Ki-Sik; An, Sung Jin; Theodorou, Christoforos G.; Ghibaudo, Gerard; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 20

2020

Effects of Contact Potential and Sidewall Surface Plane on the Performance of GaN Vertical Nanowire MOSFETs for Low-Voltage Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Son, Dong-Hyeok; Thingujam, Terirama; Kim, Jeong-Gil; Kim, Dae-Hyun; Kang, In Man; Im, Ki-Sik; Theodorou, Christoforos; Ghibaudo, Gerard; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 7

2020

Effects of Interface Traps and Self-Heating on the Performance of GAA GaN Vertical Nanowire MOSFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Thingujam, Terirama; Son, Dong-Hyeok; Kim, Jeong-Gil; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 19

2020

Subthreshold Characteristics of AlGaIn/GaN MIS-FinFETs with Controlling Threshold Voltages

ELECTRONICS

Dai, Quan; Lee, Jung-Hee

被引用次数: 2

2020

Gallium Nitride Normally Off MOSFET Using Dual-Metal-Gate Structure for the Improvement in Current Drivability

ELECTRONICS

Yoon, Young Jun; Lee, Jae Sang; Kim, Dong-Seok; Lee, Jung-Hee; Kang, In Man

被引用次数: 8

2020

Design and Analysis of Gallium Nitride-Based p-i-n Diode Structure for Betavoltaic Cell with Enhanced Output Power Density

MICROMACHINES

Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Jung Hee; Kim, Dong Seok

被引用次数: 3

2020

Effect of Thin Al₂O₃ Surface Protection Layer on Performance Improvement of AlGa_N/Ga_N HEMT

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Ryun-Hwi; Kang, Seung-Hyeon; Choi, Bong-Yeol; Lee, Jung-Hee

被引用次数: 0

2020

Deep Sub-60 mV/decade Subthreshold Swing in AlGa_N/Ga_N FinMISHFETs with M-Plane Sidewall Channel

IEEE TRANSACTIONS ON ELECTRON DEVICES

Dai, Quan; Son, Dong-Hyeok; Yoon, Young-Jun; Kim, Jeong-Gil; Jin, Xiaoshi; Kang, In-Man; Kim, Dae-Hyun; Xu, Yue; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 12

2019

Long-channel InAlAs/InGaAs/InAlAs single-quantum-well MISFETs with subthreshold swing of 61 mV/decade and effective mobility of 11 900 cm² V⁻¹ s⁻¹

APPLIED PHYSICS EXPRESS

Lee, In-Geun; Joe, Hyeon-Bhin; Yun, Do-Young; Shin, Chan-Soo; Lee, Jung-Hee; Kim, Tae-woo; Ko, Dae-Hong; Kim, Dae-Hyun

被引用次数: 2

2019

L_g=25 nm InGaAs/InAlAs high-electron mobility transistors with both f_T and f_{max} in excess of 700 GHz

APPLIED PHYSICS EXPRESS

Jo, Hyeon-Bhin; Yun, Do-Young; Baek, Ji-Min; Lee, Jung-Hee; Kim, Tae-Woo; Kim, Dae-Hyun; Tsutsumi, Takuya; Sugiyama, Hiroki; Matsuzaki, Hideaki

被引用次数: 28

2019

The 25th Korean Conference on Semiconductors Preface

SOLID-STATE ELECTRONICS

Lee, Jung-Hee; Park, Tae Joo; Kim, Kyung Rok; Kwon, Yongwoo

被引用次数: 0

2019

Growth of AlGa_N/Ga_N heterostructure with lattice-matched AlIn(Ga)_N back barrier

SOLID-STATE ELECTRONICS

Kim, Jeong-Gil; Kang, Seung-Hyeon; Janicki, Lukasz; Lee, Jun-Hyeok; Ju, Jeong-Min; Kim, Kyung-Wan; Lee, Yong-Soo; Lee, Sang-Heung; Lim, Jong-Won; Kwon, Ho-Sang; Lee, Jung-Hee

被引用次数: 10

2019

Scattering mechanisms in In_{0.7}Ga_{0.3}As/In_{0.52}Al_{0.48}As quantum-well metal-oxidesemiconductor field-effect transistors

SOLID-STATE ELECTRONICS

George, Sethu Merin; Son, Seung-Woo; Lee, Jung-Hee; Kim, Tae-Woo; Kim, Dae-Hyun

被引用次数: 1

2019

Trap and 1/f-noise effects at the surface and core of GaN nanowire gate-all-around FET structure

NANO RESEARCH

Reddy, Mallem Siva Pratap; Im, Ki-Sik; Lee, Jung-Hee; Caulmione, Raphael; Cristoloveanu, Sorin

被引用次数: 17

2019

Trichromophore-doped cassava-based biopolymer as low-cost and eco-friendly luminous material for bio hybrid white-light-emitting diodes by dual-FRET process

OPTICAL MATERIALS

Mallem, Siva Pratap Reddy; Im, Ki-Sik; Lee, Jung-Hee; Park, Chinho; Bathalavaram, Poornaprakash

被引用次数: 4

2019

Low-Frequency Noise Characteristics of GaN Nanowire Gate-All-Around Transistors With/Without 2-DEG Channel

IEEE TRANSACTIONS ON ELECTRON DEVICES

Im, Ki-Sik; Reddy, M. Siva Pratap; Caulmilone, Raphael; Theodorou, Christoforos G.; Ghibaudo, Gerard; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 18

2019

Performance of Recessed Anode AlGaIn/GaN Schottky Barrier Diode Passivated With High-Temperature Atomic Layer-Deposited Al₂O₃ Layer

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Jae-Hoon; Im, Ki-Sik; Kim, Jong Kyu; Lee, Jung-Hee

被引用次数: 11

2019

1/f noise characteristics of AlGaIn/GaN HEMTs with periodically carbon-doped GaN buffer layer

MICROELECTRONIC ENGINEERING

Im, Ki-Sik; Choi, Jinseok; Hwang, Youngmin; An, Sung Jin; Roh, Jea-Seung; Kang, Seung-Hyeon; Lee, Jun-Hyeok; Lee, Jung-Hee

被引用次数: 14

2019

Temperature- and light-sensitive mechanism in metal/organic/n-GaN bio-hybrid temperature photodiode based on salmon DNA biomolecule

JOURNAL OF MATERIALS SCIENCE-MATERIALS IN ELECTRONICS

Reddy, M. Siva Pratap; Puneetha, Peddathimula; Lee, Jung-Hee; Shim, Jaesool; Im, Ki-Sik

被引用次数: 3

2019

GaN-Based Ultraviolet Passive Pixel Sensor on Silicon (111) Substrate

SENSORS

Lee, Chang-Ju; Won, Chul-Ho; Lee, Jung-Hee; Hahm, Sung-Ho; Park, Hongsik

被引用次数: 14

2019

High Sensitive pH Sensor Based on AlInN/GaN Heterostructure Transistor

SENSORS

Dong, Yan; Son, Dong-Hyeok; Dai, Quan; Lee, Jun-Hyeok; Won, Chul-Ho; Kim, Jeong-Gil; Chen, Dunjun; Lee, Jung-Hee; Lu, Hai; Zhang, Rong; Zheng, Youdou

被引用次数: 16

2018

AlGa_N/Ga_N heterostructure pH sensor with multi-sensing segments

SENSORS AND ACTUATORS B-CHEMICAL

Dong, Yan; Son, Dong-Hyeok; Dai, Quan; Lee, Jun-Hyeok; Won, Chul-Ho; Kim, Jeong-Gil; Kang, Seung-Hyeon; Lee, Jung-Hee; Chen, Dunjun; Lu, Hai; Zhang, Rong; Zheng, Youdou

被引用次数: 45

2018

Performance Improvement and Sub-60 mV/Decade Swing in AlGa_N/Ga_N FinFETs by Simultaneous Activation of 2DEG and Sidewall MOS Channels

IEEE TRANSACTIONS ON ELECTRON DEVICES

Xu, Yue; Cristoloveanu, Sorin; Bawedin, Maryline; Im, Ki-Sik; Lee, Jung-Hee

被引用次数: 14

2018

Stepper-based L_g=0.5 μm In_{0.52}Al_{0.48}As/In_{0.7}Ga_{0.3}As PHEMTs on a 3-inch InP substrate with record product of f_T and L_g

SOLID-STATE ELECTRONICS

Baek, Ji-Min; Son, Seung-Woo; Park, Jung Ho; Park, Jong-Keun; Kwak, Jeong-Geun; Yoon, Jacoby; Bang, Dong-Soo; Lee, Jung-Hee; Kim, Taewoo; Kim, Dae-Hyun

被引用次数: 3

2018

Low voltage operation of Ga_N vertical nanowire MOSFET

SOLID-STATE ELECTRONICS

Son, Dong-Hyeok; Jo, Young-Woo; Seo, Jae Hwa; Won, Chul-Ho; Im, Ki-Sik; Lee, Yong Soo; Jang, Hwan Soo; Kim, Dae-Hyun; Kang, In Man; Lee, Jung-Hee

被引用次数: 28

2018

Normally-off AlGa_N/Ga_N-based MOS-HEMT with self-terminating TMAH wet recess etching

SOLID-STATE ELECTRONICS

Son, Dong-Hyeok; Jo, Young-Woo; Won, Chul-Ho; Lee, Jun-Hyeok; Seo, Jae Hwa; Lee, Sang-Heung; Lim, Jong-Won; Kim, Ji Heon; Kang, In Man; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 9

2018

Selected papers from The 24th Korean Conference on Semiconductors 2017 Preface

SOLID-STATE ELECTRONICS

Lee, Jung-Hee; Kim, Dae-Hwan; Min, Yo-Sep; Cho, Seong-Jae

被引用次数: 0

2018

Physical investigation of gate capacitance in In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As quantum-well metal-oxide-semiconductor field-effect-transistors

AIP ADVANCES

Jo, Hyeon-Bhin; Park, Jung-Ho; Son, Seung-Woo; Baek, Ji-Min; Yun, Do-Young; Kang, Yeonsung; Seo, Yong-Hyun; Yung, Lee Min; Lee, Jung-Hee; Kim, Tae-Woo; Kim, Dae-Hyun

被引用次数: 0

2018

Residue-and-polymer-free graphene transfer: DNA-CTMA/graphene/GaN bio-hybrid photodiode for light-sensitive applications

OPTICAL MATERIALS

Reddy, M. Siva Pratap; Park, Herie; Lee, Jung-Hee

被引用次数: 19

2018

Temperature-dependent electrical characteristics and carrier transport mechanism of p-Cu₂ZnSnS₄/n-GaN heterojunctions

APPLIED PHYSICS A-MATERIALS SCIENCE & PROCESSING

Reddy, Varra Niteesh; Reddy, M. Siva Pratap; Gunasekhar, K. R.; Lee, Jung-Hee

被引用次数: 5

2018

Bufferless GaN-Based MOSFETs Fabricated on GaN-on-Insulator Wafer

PHYSICA STATUS SOLIDI A-APPLICATIONS AND MATERIALS SCIENCE

Won, Chul-Ho; Caulmilone, Raphael; Cristoloveanu, Sorin; Im, Ki-Sik; Lee, Jung-Hee

被引用次数: 0

2018

Impact of the Source-to-Drain Spacing on the DC and RF Characteristics of InGaAs/InAlAs High-Electron Mobility Transistors

IEEE ELECTRON DEVICE LETTERS

Yun, Do-Young; Jo, Hyeon-Bhin; Son, Seung-Woo; Baek, Ji-Min; Lee, Jung-Hee; Kim, Tae-Woo; Kim, Dae-Hyun; Tsutsumi, Takuya; Sugiyama, Hiroki; Matsuzaki, Hideaki

被引用次数: 15

2018

L_g=87 nm InAlAs/InGaAs High-Electron-Mobility Transistors With a gm_{max} of 3 S/mm and f_T of 559 GHz

IEEE ELECTRON DEVICE LETTERS

Jo, Hyeon-Bhin; Baek, Ji-Min; Yun, Do-Young; Son, Seung-Woo; Lee, Jung-Hee; Kim, Tae-Woo; Kim, Dae-Hyun; Tsutsumi, Takuya; Sugiyama, Hiroki; Matsuzaki, Hideaki

被引用次数: 35

2018

A Novel Analysis of L_{gd} Dependent-1/f Noise in In_{0.08}Al_{0.92}N/GaN

IEEE ELECTRON DEVICE LETTERS

Seo, Jae Hwa; Yoon, Young Jun; Son, Dong-Hyeok; Kim, Jeong-Gil; Lee, Jong-Ho; Lee, Jung-Hee; Im, Ki-Sik; Kang, In Man

被引用次数: 4

2018

Total-Ionizing-Dose Responses of GaN-Based HEMTs With Different Channel Thicknesses and MOSHEMTs With Epitaxial MgCaO as Gate Dielectric

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Bhuiyan, Maruf A.; Zhou, Hong; Chang, Sung-Jae; Lou, Xiabing; Gong, Xian; Jiang, Rong; Gong, Huiqi; Zhang, En Xia; Won, Chul-Ho; Lim, Jong-Won; Lee, Jung-Hee; Gordon, Roy G.; Reed, Robert A.; Fleetwood, Daniel M.; Ye, Peide; Ma, Tso-Ping

被引用次数: 18

2018

Proton Irradiation Effects on AlGa_N/Ga_N HEMTs With Different Isolation Methods

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Kim, Dong-Seok; Lee, Jun-Hyeok; Yeo, Sunmog; Lee, Jung-Hee

被引用次数: 21

2018

High Figure-of-Merit(VBR₂/RON) AlGa_N/Ga_N Power HEMT With Periodically C-Doped Ga_N Buffer and AlGa_N Back Barrier

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Lee, Jun-Hyeok; Ju, Jeong-Min; Atmaca, Gokhan; Kim, Jeong-Gil; Kang, Seung-Hyeon; Lee, Yong Soo; Lee, Sang-Heung; Lim, Jong-Won; Kwon, Ho-Sang; Lisesivdin, Sefer Bora; Lee, Jung-Hee

被引用次数: 33

2018

Dual-Surface Modification of AlGa_N/Ga_N HEMTs Using TMAH and Piranha Solutions for Enhancing Current and 1/f-Noise Characteristics

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Reddy, M. Siva Pratap; Park, Won-Sang; Im, Ki-Sik; Lee, Jung-Hee

被引用次数: 12

2018

Current Collapse-Free and Self-Heating Performances in Normally Off Ga_N Nanowire GAA-MOSFETs

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Im, Ki-Sik; Atmaca, Gokhan; Won, Chul-Ho; Caulmilone, Raphael; Cristoloveanu, Sorin; Kim, Yong-Tae; Lee, Jung-Hee

被引用次数: 6

2018

Performance enhancement of AlGa_N/Ga_N nanochannel omega-FinFET

SOLID-STATE ELECTRONICS

Im, Ki-Sik; Seo, Jae Hwa; Vodapally, Sindhuri; Kang, In Man; Lee, Jae-Hoon; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 4

2017

Growth of 10 nm-thick AlIn(Ga)_N/Ga_N heterostructure with high electron mobility and low sheet resistance

PHYSICA STATUS SOLIDI B-BASIC SOLID STATE PHYSICS

Kim, Jeong-Gil; Im, Ki-Sik; Won, Chul-Ho; Kang, Seung-Hyeon; Lee, Sang-Heung; Lim, Jong-Won; Kim, Ji Heon; Lee, Jung-Hee

被引用次数: 3

2017

The effects of tetramethylammonium hydroxide treatment on the performance of recessed-gate AlGa_N/Ga_N high electron mobility transistors

THIN SOLID FILMS

Do, Jae-Won; Jung, Hyun-Wook; Shin, Min Jeong; Ahn, Ho-Kyun; Kim, Haecheon; Kim, Ryun-Hwi; Cho, Kyu Jun; Chang, Sung-Jae; Min, Byoung-Gue; Yoon, Hyung Sup; Kim, Ji-Heon; Yang, Jin-Mo; Lee, Jung-Hee; Lim, Jong-Won

被引用次数: 9

2017

Hydrazine (N₂H₄)-Based Surface Treatment for Interface Quality Improvement in Al₂O₃/AlGa_N/Ga_N MIS-HEMT

ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY

Jung, Hyun-Wook; Shin, Min Jeong; Chang, Sung-Jae; Ahn, Ho-Kyun; Do, Jae-Won; Cho, Kyu Jun; Won, Chul-Ho; Min, Byoung-Gue; Kim, Haecheon; Yoon, Hyung Sup; Lee, Jung-Hee; Lim, Jong-Won

被引用次数: 1

2017

Record Effective Mobility Obtained From In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As Quantum-Well MOSFETs on 300-mm Si Substrate

IEEE ELECTRON DEVICE LETTERS

Son, Seung-Woo; Park, Jung Ho; Baek, Ji-Min; Kim, Do-Kwyn; Lee, Seung Ryul; Lee, Sang-Moon; Yoon, Jieon; Kim, Jungtaek; Song, Woo-Bin; Kim, Sunjung; Shin, Dong Suk; Kim, Yihwan; Shin, Seung Heon; Kim, Tae-Woo; Lee, Jung-Hee; Kim, Dae-Hyun

被引用次数: 9

2017

1/f-Noise in AlGa_N/Ga_N Nanowire Omega-FinFETs

IEEE ELECTRON DEVICE LETTERS

Vodapally, Sindhuri; Jang, Young In; Kang, In Man; Cho, In-Tak; Lee, Jong-Ho; Bae, Youngho; Ghibaudo, Gerard; Cristoloveanu, Sorin; Im, Ki-Sik; Lee, Jung-Hee

被引用次数: 27

2017

Comparison for 1/f Noise Characteristics of AlGa_N/Ga_N FinFET and Planar MISHFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Vodapally, Sindhuri; Theodorou, Christoforos G.; Bae, Youngho; Ghibaudo, Gerard; Cristoloveanu, Sorin; Im, Ki-Sik; Lee, Jung-Hee

被引用次数: 33

2017

Capacitance-voltage characterization of Al₂O₃/Ga_N-on-insulator (GaNOI) structures with TMAH surface treatment

MICROELECTRONIC ENGINEERING

Im, Ki-Sik; Kim, Jeong-Gil; Vodapally, Sindhuri; Caulmilone, Raphael; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 4

2017

A novel high-performance H-gate U-channel junctionless FET

JOURNAL OF COMPUTATIONAL ELECTRONICS

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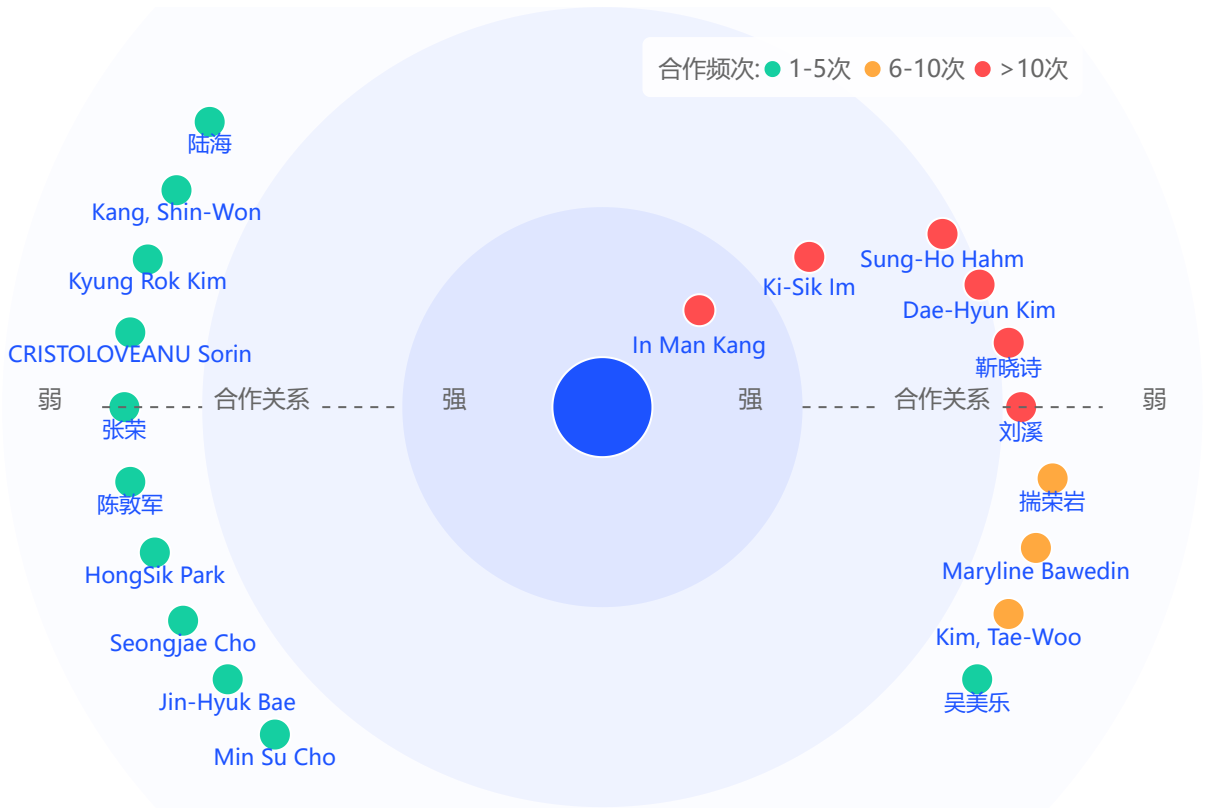
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学术合作



In Man Kang	合作次数: 49
庆北国立大学	
Ki-Sik Im	合作次数: 33
庆北国立大学	
Sung-Ho Hahm	合作次数: 16
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